

CUSUM Active-Manager Monitor

Statistical Process Control for Portfolio Manager Oversight

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What is CUSUM?

CUSUM (Cumulative Sum) is a sequential change-point detection method from statistical process control, adapted for investment management by Philips, Yashchin & Stein (2003). It answers one critical question:

Has a fund manager lost their investment skill — and if so, exactly when?

The Case for CUSUM — Why CIOs Should Care

Cost of Inaction:

Retaining an underperforming manager for 3 extra years on a 500M mandate at IR=-0.3, TE=6% costs ~9M in excess losses. Early, systematic detection pays.

Faster than Rolling IR:

A trailing 3-year IR review needs 10+ years of data to distinguish skill from luck at 95% confidence. CUSUM detects persistent CHANGES in IR within 2-3 years.

Rigorous False-Alarm Control:

Threshold h is calibrated from the paper's ARL table: $h=19.81$ gives ~1 false alarm per 60 months (5 years) when the true IR=+0.5. Fewer spurious terminations.

Two-Sided Monitoring:

Detects skill LOSS (lower CUSUM) and unexpected skill GAIN (upper CUSUM), enabling both firing decisions and early-stage manager promotions.

Rolling Restart:

After each alarm L resets to zero — monitoring never stops. Multiple alarms build a richer signal; post-alarm returns can be tracked to validate each flag.

Industry Precedent:

SPC methods are standard in manufacturing, clinical trials, and nuclear operations. Mercer and WTW publish SPC-adjacent monitoring frameworks for manager oversight.

Actionable, Not Automatic:

An alarm triggers a structured review — not automatic termination. It forces the investment committee to ask: Is the alpha source still intact?

How CUSUM Works

Step-by-step mechanics of the Philips-Yashchin-Stein procedure

1

Step 1 — Log Excess Return

Each month: $e_t = \ln[(1 + r_{\text{fund}}) / (1 + r_{\text{benchmark}})]$

Log returns are additive and symmetric — they handle compounding correctly.

2

Step 2 — EWMA Tracking-Error Estimate

$\sigma^2_t = \gamma * \sigma^2_{t-1} + (1-\gamma) * 0.5*(e_t - e_{t-1})^2$ [gamma=0.90]

Von Neumann adjacent-difference estimator: robust to drift in the mean, adapts as the manager's risk budget changes.

3

Step 3 — Monthly Information Ratio Estimate

$IR_{\text{hat}_t} = (e_t * \sqrt{12}) / \sigma_{t-1}$

Annualised IR using lagged sigma to keep the estimator approximately unbiased.

4

Step 4 — Lindley Recursion (Page's CUSUM)

$L_t = \max(0, L_{t-1} + \text{drift})$ $\text{drift} = 0.5*(IR_{\text{good}} + IR_{\text{bad}}) - IR_{\text{hat}_t}$

Default: $IR_{\text{good}}=0.5$, $IR_{\text{bad}}=0.0$ -> $\text{drift} = 0.25 - IR_{\text{hat}}$. L grows when manager underperforms, resets to 0 when they outperform. Upper CUSUM L_{up} uses $\text{drift}_{\text{up}} = IR_{\text{hat}} - 0.5*(IR_{\text{exceptional}} + IR_{\text{good}})$.

5

Step 5 — Rolling Alarm Rule

ALARM if $L_t \geq h$ [default $h = 19.81$] then reset $L_t = 0$ and continue

After each alarm L resets to zero — monitoring never stops. All alarm times are recorded; post-alarm returns validate each signal.

Alarm Threshold Reference Table (Table 2, Philips et al. 2003)

Threshold (h)	ARL — Good (IR=0.5)	ARL — Flat (IR=0)	ARL — Bad (IR=-0.5)
11.81	24.00	16.00	11.00
15.00	36.00	22.00	15.00
17.60	48.00	27.00	18.00
19.81	60.00	32.00	21.00
21.79	72.00	37.00	23.00
23.59	84.00	41.00	25.00

Fund Universe

69 active managers — page 1 of 4

#	Ticker	Fund Name	Benchmark	Asset Class	Start	h
1	FCNTX	Fidelity Contrafund vs Russell 1000 Growth	IWF	US Large Growth	Jan 2005	19.81
2	DODGX	Dodge & Cox Stock vs Russell 1000 Value	IWD	US Large Value	Jan 2005	19.81
3	AGTHX	AmFunds Growth Fund of America vs S&P 500	SPY	US Large Blend	Jan 2005	19.81
4	OTCFX	T. Rowe Price Small-Cap Stock vs Russell 2000	IWM	US Small Cap	Jan 2005	19.81
5	OAKIX	Oakmark International vs MSCI EAFE	EFA	Intl Developed	Jan 2005	19.81
6	TEDMX	Templeton Developing Markets vs MSCI EM	EEM	Emerging Markets	Jan 2005	19.81
7	VWELX	Vanguard Wellington vs 60/40 Allocation	AOR	Balanced 60/40	Jan 2005	19.81
8	PTTRX	PIMCO Total Return vs US Agg Bond	AGG	US Fixed Income	Jan 2005	19.81
9	PRWCX	T. Rowe Price Capital Appreciation vs 60/40 Allocation	AOR	Balanced 60/40	Jan 2005	19.81
10	FMAGX	Fidelity Magellan vs S&P 500	SPY	US Large Blend	Jan 2005	19.81
11	SEQUX	Sequoia Fund vs S&P 500	SPY	US Large Blend	Jan 2005	19.81
12	FLPSX	Fidelity Low-Priced Stock vs Russell 2000	IWM	US Small/Mid Cap	Jan 2005	19.81
13	DODWX	Dodge & Cox International vs MSCI EAFE	EFA	Intl Value	Jan 2005	19.81
14	LSBRX	Loomis Sayles Bond vs US Agg Bond	AGG	US Fixed Income	Jan 2005	19.81
15	NWFFX	New World Fund vs MSCI EM	EEM	Emerging Markets	Jan 2005	19.81
16	PRMSX	T. Rowe Price EM Stock vs MSCI EM	EEM	Emerging Markets	Jan 2005	19.81
17	JEMWX	JPMorgan EM Equity vs MSCI EM	EEM	Emerging Markets	Jan 2005	19.81
18	ODMAX	Invesco Developing Markets vs MSCI EM	EEM	Emerging Markets	Jan 2005	19.81
19	PEIFX	PGIM Jennison EM Equity vs MSCI EM	EEM	Emerging Markets	Jan 2005	19.81
20	LZOEX	Lazard EM Equity vs MSCI EM	EEM	Emerging Markets	Jan 2005	19.81

US Equity (FCNTX, DODGX, AGTHX, OTCFX, FMAGX, SEQUX, FLPSX)
Intl & Emerging Markets (OAKIX, TEDMX, DODWX, + 26 EM equity funds)
Fixed Income / Balanced (VWELX, PTTRX, PRWCX, LSBRX)
EM Debt: Hard CCY vs EMB (PEBIX, EBNDX, PREMX ...) & Local CCY vs EMLC
Global Equity vs ACWI / URTH (CWGIX, VHGX, AIVSX ... + AIAIM funds)

Fund Universe

69 active managers — page 2 of 4

#	Ticker	Fund Name	Benchmark	Asset Class	Start	h
21	HLMEX	Harding Loevner EM Equity vs MSCI EM	EEM	Emerging Markets	Jan 2005	19.81
22	MEMCX	Matthews EM vs MSCI EM	EEM	Emerging Markets	Jan 2005	19.81
23	VEIEX	Vanguard EM Stock Index vs MSCI EM	EEM	Emerging Markets	Jan 2005	19.81
24	VEMAX	Vanguard EM Stock Index Admiral vs MSCI EM	EEM	Emerging Markets	Jan 2005	19.81
25	NEWFX	American Funds New World vs MSCI EM	EEM	Emerging Markets	Jan 2005	19.81
26	HAEMX	Harbor EM vs MSCI EM	EEM	Emerging Markets	Jan 2005	19.81
27	TEMTX	Templeton EM vs MSCI EM	EEM	Emerging Markets	Jan 2005	19.81
28	SFENX	Seafarer EM Growth & Income vs MSCI EM	EEM	Emerging Markets	Jan 2005	19.81
29	REMITX	Royce EM vs MSCI EM	EEM	Emerging Markets	Jan 2005	19.81
30	ABEMX	AB EM Growth vs MSCI EM	EEM	Emerging Markets	Jan 2005	19.81
31	GERIX	Goldman Sachs EM Equity vs MSCI EM	EEM	Emerging Markets	Jan 2005	19.81
32	HLEMX	Harding Loevner EM Institutional vs MSCI EM	EEM	Emerging Markets	Jan 2005	19.81
33	BGEFX	Baron EM vs MSCI EM	EEM	Emerging Markets	Jan 2005	19.81
34	WIEMX	William Blair EM Growth vs MSCI EM	EEM	Emerging Markets	Jan 2005	19.81
35	WAEMX	Western Asset EM vs MSCI EM	EEM	Emerging Markets	Jan 2005	19.81
36	MIEMX	MFS EM Equity vs MSCI EM	EEM	Emerging Markets	Jan 2005	19.81
37	EEMAX	Eaton Vance EM vs MSCI EM	EEM	Emerging Markets	Jan 2005	19.81
38	FSEAX	Fidelity EM vs MSCI EM	EEM	Emerging Markets	Jan 2005	19.81
39	MADCX	Madison EM vs MSCI EM	EEM	Emerging Markets	Jan 2005	19.81
40	PDEZX	PGIM EM vs MSCI EM	EEM	Emerging Markets	Jan 2005	19.81

Fund Universe

69 active managers — page 3 of 4

#	Ticker	Fund Name	Benchmark	Asset Class	Start	h
41	PEBIX	PIMCO EM Bond vs JPM EM Bond (Hard CCY)	EMB	EM Hard CCY Debt	Jan 2005	19.81
42	EBNDX	BlackRock EM Bond vs JPM EM Bond (Hard CCY)	EMB	EM Hard CCY Debt	Jan 2005	19.81
43	PREMX	T. Rowe Price EM Bond vs JPM EM Bond (Hard CCY)	EMB	EM Hard CCY Debt	Jan 2005	19.81
44	ABEMX	AB EM Debt vs JPM EM Bond (Hard CCY)	EMB	EM Hard CCY Debt	Jan 2005	19.81
45	TGBAX	Templeton Global Bond vs JPM EM Bond (Hard CCY)	EMB	EM Hard CCY Debt	Jan 2005	19.81
46	IEMIX	Invesco EM Sovereign Debt vs JPM EM Bond (Hard CCY)	EMB	EM Hard CCY Debt	Jan 2005	19.81
47	DBLEX	DFA EM Fixed Income vs JPM EM Bond (Hard CCY)	EMB	EM Hard CCY Debt	Jan 2005	19.81
48	VWOB	Vanguard EM Govt Bond ETF vs JPM EM Bond (Hard CCY)	EMB	EM Hard CCY Debt	Jan 2005	19.81
49	MSEDX	MFS EM Debt vs JPM EM Bond (Hard CCY)	EMB	EM Hard CCY Debt	Jan 2005	19.81
50	PFMIX	PIMCO Foreign Bond vs VanEck EM Local CCY Bond	EMLC	EM Local CCY Debt	Jan 2005	19.81
51	PRLAX	T. Rowe Price EM Local CCY Bond vs VanEck EM Local CCY Bond	EMLC	EM Local CCY Debt	Jan 2005	19.81
52	CWGIX	Capital World Growth & Income vs MSCI ACWI	ACWI	Global Equity	Jan 2005	19.81
53	TEDIX	Templeton Global Equity vs MSCI ACWI	ACWI	Global Equity	Jan 2005	19.81
54	VHGEX	Vanguard Global Equity vs MSCI ACWI	ACWI	Global Equity	Jan 2005	19.81
55	OPGIX	Invesco Oppenheimer Global vs MSCI ACWI	ACWI	Global Equity	Jan 2005	19.81
56	GPROX	Goldman Sachs Global Equity vs MSCI ACWI	ACWI	Global Equity	Jan 2005	19.81
57	SGIGX	Schroders Global vs MSCI ACWI	ACWI	Global Equity	Jan 2005	19.81
58	MGGIX	MFS Global Equity vs MSCI ACWI	ACWI	Global Equity	Jan 2005	19.81
59	AIVSX	American Investors vs MSCI World	URTH	Global Equity	Jan 2005	19.81
60	MGIAX	Morgan Stanley Institutional Global vs MSCI World	URTH	Global Equity	Jan 2005	19.81

Fund Universe

69 active managers — page 4 of 4

#	Ticker	Fund Name	Benchmark	Asset Class	Start	h
61	JGVAX	JPMorgan Global Focus vs MSCI World	URTH	Global Equity	Jan 2005	19.81
62	HAINX	Harbor International vs MSCI World	URTH	Global Equity	Jan 2005	19.81
63	0P0001I6KI.SI	AIAIM Global Equity Fund A vs MSCI ACWI	ACWI	Global Equity	Jan 2005	19.81
64	0P0001K2U1.SI	AIAIM Global Equity Fund B vs MSCI ACWI	ACWI	Global Equity	Jan 2005	19.81
65	0P00008T6G.SI	AIAIM Global Equity Fund C vs MSCI ACWI	ACWI	Global Equity	Jan 2005	19.81
66	0P00008T6D.SI	AIAIM Global Equity Fund D vs MSCI ACWI	ACWI	Global Equity	Jan 2005	19.81
67	0P0001RMIN.SI	AIAIM Global Equity Fund E vs MSCI ACWI	ACWI	Global Equity	Jan 2005	19.81
68	0P0001KL95.SI	AIAIM EM Bond Fund A vs JPM EM Bond	EMB	EM Hard CCY Debt	Jan 2005	19.81
69	0P00008T6N.SI	AIAIM EM Bond Fund B vs JPM EM Bond	EMB	EM Hard CCY Debt	Jan 2005	19.81

Summary Results

CUSUM alarm status and key statistics (Jan 2005 - present) — page 1 of 4

Fund	Bench	Months	Ann TE	IR (95% CI)	Cum Excess	# Alarms	Alarm?	First Alarm
FCNTX	IWF	256	4.3%	-0.00 [-0.39,+0.38]	+3.4%	9	YES	Feb 2010
DODGX	IWD	256	4.4%	+0.21 [-0.24,+0.67]	+14.5%	7	YES	Mar 2008
AGTHX	SPY	256	4.6%	+0.13 [-0.33,+0.63]	+9.9%	9	YES	Oct 2008
OTCFX	IWM	256	4.1%	+0.24 [-0.22,+0.75]	+32.7%	6	YES	Nov 2007
OAKIX	EFA	256	8.2%	+0.14 [-0.30,+0.57]	+11.0%	8	YES	Sep 2007
TEDMX	EEM	256	5.9%	+0.03 [-0.31,+0.41]	+2.7%	5	YES	Jan 2007
VWELX	AOR	210	4.0%	+0.65 [+0.39,+1.10]	+58.0%	0	NO	-
PTTRX	AGG	256	1.9%	+0.50 [+0.03,+1.03]	+22.2%	5	YES	Dec 2008
PRWCX	AOR	210	4.6%	+0.99 [+0.67,+1.45]	+119.1%	1	YES	Jan 2026
FMAGX	SPY	256	5.7%	-0.15 [-0.66,+0.40]	-24.0%	9	YES	Sep 2006
SEQUX	SPY	256	8.4%	-0.22 [-0.64,+0.23]	-33.2%	9	YES	Apr 2007
FLPSX	IWM	256	8.3%	+0.00 [-0.38,+0.37]	+17.2%	6	YES	Aug 2008
DODWX	EFA	216	7.1%	+0.48 [+0.05,+0.96]	+72.3%	3	YES	Jan 2016
LSBRX	AGG	256	6.9%	+0.26 [-0.25,+0.88]	+39.1%	7	YES	Jul 2008
NWFFX	EEM	256	7.8%	+0.06 [-0.30,+0.43]	+29.7%	7	YES	Jan 2006
PRMSX	EEM	256	4.6%	-0.06 [-0.56,+0.41]	-9.6%	6	YES	Sep 2008
JEMWX	EEM	149	5.0%	+0.42 [-0.21,+1.15]	+27.3%	2	YES	Jan 2022
ODMAX	EEM	256	5.9%	+0.06 [-0.33,+0.43]	+14.2%	5	YES	Apr 2015

Summary Results

CUSUM alarm status and key statistics (Jan 2005 - present) — page 2 of 4

Fund	Bench	Months	Ann TE	IR (95% CI)	Cum Excess	# Alarms	Alarm?	First Alarm
PEIFX	EEM	131	8.6%	+0.25 [-0.47,+0.91]	+22.9%	4	YES	Apr 2019
LZOEX	EEM	256	6.3%	+0.14 [-0.29,+0.56]	+23.4%	4	YES	Mar 2015
HLMEX	EEM	247	5.7%	-0.09 [-0.45,+0.30]	-7.1%	6	YES	Oct 2008
MEMCX	EEM	256	4.6%	-0.36 [-0.72,-0.04]	-26.9%	8	YES	Sep 2007
VEIEX	EEM	256	4.1%	-0.12 [-0.46,+0.29]	-6.8%	5	YES	Oct 2008
VEMAX	EEM	239	3.7%	-0.06 [-0.44,+0.35]	-1.5%	5	YES	Oct 2008
NEWFX	EEM	256	7.8%	+0.06 [-0.30,+0.43]	+29.4%	7	YES	Jan 2006
HAEMX	EEM	47	20.8%	-0.49 [-1.03,+1.09]	-39.3%	2	YES	Mar 2020
TEMTX	EEM	256	10.5%	-0.23 [-0.68,+0.19]	-33.1%	9	YES	Jan 2006
SFENX	EEM	220	6.6%	+0.08 [-0.39,+0.54]	+10.0%	6	YES	Jun 2012
REMITX	EEM	76	17.5%	+0.04 [-0.74,+0.84]	+13.1%	2	YES	Mar 2025
ABEMX	EEM	228	5.9%	+0.20 [-0.24,+0.65]	+30.8%	5	YES	Dec 2013
GERIX	EEM	223	4.5%	+0.27 [-0.15,+0.67]	+21.5%	3	YES	Oct 2008
HLEMTX	EEM	83	5.7%	-0.38 [-1.02,+0.36]	-15.7%	3	YES	Sep 2020
BGEFTX	EEM	133	12.2%	+0.10 [-0.58,+0.80]	+6.9%	5	YES	Aug 2016
WIEMTX	EEM	123	12.0%	-0.06 [-0.65,+0.57]	-10.2%	5	YES	Jan 2017
WAEMTX	EEM	224	11.3%	+0.18 [-0.28,+0.63]	+40.9%	7	YES	Oct 2008
MIEMTX	EEM	158	7.2%	-0.03 [-0.50,+0.45]	+2.3%	5	YES	Apr 2011

Summary Results

CUSUM alarm status and key statistics (Jan 2005 - present) — page 3 of 4

Fund	Bench	Months	Ann TE	IR (95% CI)	Cum Excess	# Alarms	Alarm?	First Alarm
EEMAX	EEM	224	5.5%	+0.08 [-0.42,+0.59]	+4.7%	5	YES	Sep 2008
FSEAX	EEM	256	8.9%	+0.41 [-0.08,+0.93]	+120.3%	4	YES	Apr 2008
MADCX	EEM	256	5.3%	+0.13 [-0.22,+0.49]	+14.7%	5	YES	Sep 2008
PDEZX	EEM	140	11.1%	+0.23 [-0.41,+0.86]	+27.6%	4	YES	Mar 2017
PEBIX	EMB	221	4.5%	+0.07 [-0.24,+0.55]	+9.1%	4	YES	Jun 2012
EBNDX	EMB	133	19.1%	-0.07 [-0.67,+0.57]	-28.9%	4	YES	Nov 2008
PREMX	EMB	221	4.9%	-0.03 [-0.35,+0.45]	-0.5%	5	YES	Sep 2011
ABEMX	EMB	221	14.3%	+0.16 [-0.35,+0.68]	+20.5%	7	YES	Nov 2008
TGBAX	EMB	221	9.5%	-0.20 [-0.67,+0.18]	-25.7%	8	YES	Sep 2011
IEMIX	EMB	4	31.8%	+1.73 [-0.61,+6.49]	+17.4%	0	NO	-
DBLEX	EMB	193	4.3%	-0.06 [-0.51,+0.36]	-2.3%	5	YES	Oct 2010
VWOB	EMB	155	1.4%	+0.04 [-0.42,+0.45]	+1.5%	4	YES	Jan 2015
MSEDX	EMB	40	4.4%	-1.42 [-2.34,-0.61]	-18.6%	3	YES	Aug 2019
PFMIX	EMLC	190	9.8%	+0.22 [-0.23,+0.62]	+51.4%	2	YES	May 2017
PRLAX	EMLC	190	17.2%	+0.19 [-0.25,+0.62]	+12.1%	5	YES	Aug 2011
CWGIX	ACWI	218	3.0%	-0.14 [-0.56,+0.27]	-5.0%	6	YES	May 2010
TEDIX	ACWI	218	8.5%	-0.21 [-0.78,+0.33]	-22.3%	8	YES	Nov 2009
VHGEX	ACWI	218	3.2%	-0.08 [-0.57,+0.39]	-7.8%	5	YES	Oct 2008

Summary Results

CUSUM alarm status and key statistics (Jan 2005 - present) — page 4 of 4

Fund	Bench	Months	Ann TE	IR (95% CI)	Cum Excess	# Alarms	Alarm?	First Alarm
OPGIX	ACWI	218	11.8%	+0.01 [-0.48,+0.52]	-16.6%	8	YES	Oct 2012
GPROX	ACWI	155	7.1%	-0.29 [-0.95,+0.36]	-27.8%	6	YES	Oct 2018
SGIGX	ACWI	101	5.0%	-0.42 [-1.17,+0.32]	-15.9%	5	YES	Mar 2020
MGGIX	ACWI	215	9.7%	+0.46 [-0.20,+1.16]	+94.9%	7	YES	Dec 2008
AIVSX	URTH	172	3.6%	+0.47 [+0.00,+0.94]	+28.5%	2	YES	May 2019
MGIAX	URTH	172	7.2%	-0.20 [-0.66,+0.20]	-16.7%	4	YES	Aug 2018
JGVAX	URTH	27	27.2%	-1.21 [-3.27,-0.93]	-55.9%	3	YES	Dec 2019
HAINX	URTH	172	6.8%	-0.72 [-1.29,-0.24]	-50.9%	10	YES	May 2013
OP0001I6KI.SACWI	ACWI	46	6.3%	-1.19 [-2.03,-0.41]	-24.0%	3	YES	Nov 2023
OP00008T6G.SACWI	ACWI	64	10.2%	-0.56 [-1.49,+0.25]	-24.9%	3	YES	Jan 2023
OP00008T6D.SACWI	ACWI	64	13.9%	+0.72 [-0.11,+1.65]	+59.2%	0	NO	-
OP0001RMIN.SACWI	ACWI	31	8.3%	-1.50 [-2.90,-0.39]	-26.6%	2	YES	Aug 2024
OP0001KL95.SMB	SMB	41	5.9%	-0.09 [-1.16,+0.96]	-1.3%	1	YES	Jan 2026
OP00008T6N.SMB	SMB	64	5.5%	-0.57 [-1.27,+0.08]	-14.3%	3	YES	Jan 2023

Post-Alarm Analysis

Cumulative excess return vs benchmark in the months following the FIRST alarm — page 1 of 5

Fund	First Alarm	+12M Fund	+12M Bench	+12M Excess	+24M Excess	+36M Excess	12M Signal
FCNTX	Feb 2010	+24.8%	+24.9%	-0.1%	-2.1%	-0.8%	CORRECT
DODGX	Mar 2008	-45.2%	-42.7%	-2.5%	+2.4%	+0.7%	CORRECT
AGTHX	Oct 2008	+15.6%	+9.7%	+5.9%	+5.0%	-2.1%	FALSE
OTCFX	Nov 2007	-37.5%	-37.6%	+0.0%	+7.5%	+14.0%	FALSE
OAKIX	Sep 2007	-28.9%	-28.8%	-0.1%	+11.9%	+20.3%	CORRECT
TEDMX	Jan 2007	+10.1%	+21.3%	-11.2%	-7.2%	-9.1%	CORRECT
PTTRX	Dec 2008	+13.9%	+3.0%	+10.9%	+14.1%	+10.9%	FALSE
PRWCX	Jan 2026	n/a	n/a	n/a	n/a	n/a	FALSE
FMAGX	Sep 2006	+22.8%	+16.4%	+6.4%	-3.7%	+1.3%	FALSE
SEQUX	Apr 2007	+3.1%	-5.0%	+8.1%	+18.2%	+12.9%	FALSE
FLPSX	Aug 2008	-11.0%	-21.3%	+10.3%	+12.8%	+14.2%	FALSE
DODWX	Jan 2016	+31.9%	+10.8%	+21.1%	+22.8%	+23.0%	FALSE
LSBRX	Jul 2008	-0.8%	+7.3%	-8.0%	+1.6%	+12.6%	CORRECT
NWFFX	Jan 2006	+24.2%	+15.0%	+9.1%	+11.9%	+10.8%	FALSE

Post-Alarm Analysis

Cumulative excess return vs benchmark in the months following the FIRST alarm — page 2 of 5

Fund	First Alarm	+12M Fund	+12M Bench	+12M Excess	+24M Excess	+36M Excess	12M Signal
PRMSX	Sep 2008	+12.8%	+16.4%	-3.6%	+0.4%	+3.2%	CORRECT
JEMWX	Jan 2022	-13.9%	-13.3%	-0.6%	-2.3%	-4.4%	CORRECT
ODMAX	Apr 2015	-14.1%	-17.9%	+3.8%	+4.5%	+6.0%	FALSE
PEIFX	Apr 2019	-23.0%	-14.2%	-8.8%	-4.4%	+9.4%	CORRECT
LZOEX	Mar 2015	-11.7%	-12.7%	+0.9%	+5.4%	-0.8%	FALSE
HMEX	Oct 2008	+48.9%	+51.0%	-2.1%	+0.4%	+0.9%	CORRECT
MEMCX	Sep 2007	-33.7%	-29.7%	-3.9%	-8.5%	-7.4%	CORRECT
VEIEX	Oct 2008	+56.0%	+51.0%	+4.9%	+4.7%	+5.6%	FALSE
VEMAX	Oct 2008	+60.2%	+51.0%	+9.2%	+12.1%	+12.5%	FALSE
NEWFX	Jan 2006	+24.2%	+15.0%	+9.1%	+11.9%	+10.8%	FALSE
HAEMX	Mar 2020	+35.1%	+58.8%	-23.7%	-54.4%	n/a	CORRECT
TEAFX	Jan 2006	+15.0%	+15.0%	-0.0%	-18.9%	-5.6%	CORRECT
SFENX	Jun 2012	+2.3%	+0.3%	+2.0%	+5.0%	-2.7%	FALSE
REMX	Mar 2025	+16.8%	+33.1%	-16.3%	n/a	n/a	CORRECT

Post-Alarm Analysis

Cumulative excess return vs benchmark in the months following the FIRST alarm — page 3 of 5

Fund	First Alarm	+12M Fund	+12M Bench	+12M Excess	+24M Excess	+36M Excess	12M Signal
ABEMX	Dec 2013	-2.4%	-3.9%	+1.5%	+3.6%	+4.9%	FALSE
GERIX	Oct 2008	+62.3%	+51.0%	+11.3%	+17.2%	+19.0%	FALSE
HLEMX	Sep 2020	+21.0%	+16.0%	+5.0%	-4.8%	-5.5%	FALSE
BGEFX	Aug 2016	+45.1%	+24.7%	+20.4%	+38.9%	+36.6%	FALSE
WIEMX	Jan 2017	+36.2%	+39.4%	-3.2%	-8.3%	+21.2%	CORRECT
WAEMX	Oct 2008	+83.0%	+51.0%	+31.9%	+97.5%	+103.4%	FALSE
MIEMX	Apr 2011	-14.5%	-13.9%	-0.6%	+3.1%	-0.2%	CORRECT
EEMAX	Sep 2008	+15.7%	+16.4%	-0.7%	+6.0%	+6.9%	CORRECT
FSEAX	Apr 2008	-42.3%	-39.8%	-2.5%	-9.7%	-6.5%	CORRECT
MADCX	Sep 2008	+23.3%	+16.4%	+6.9%	+5.5%	+9.9%	FALSE
PDEZX	Mar 2017	+33.8%	+25.0%	+8.8%	+10.1%	+21.2%	FALSE
PEBIX	Jun 2012	+1.1%	-0.8%	+1.9%	+2.2%	-2.3%	FALSE
EBNDX	Nov 2008	+64.3%	+35.5%	+28.8%	+68.3%	+43.9%	FALSE
PREMX	Sep 2011	+21.3%	+20.6%	+0.8%	+3.7%	+1.9%	FALSE

Post-Alarm Analysis

Cumulative excess return vs benchmark in the months following the FIRST alarm — page 4 of 5

Fund	First Alarm	+12M Fund	+12M Bench	+12M Excess	+24M Excess	+36M Excess	12M Signal
ABEMX	Nov 2008	+76.4%	+35.5%	+40.9%	+65.3%	+48.1%	FALSE
TGBAX	Sep 2011	+13.3%	+20.6%	-7.3%	+4.5%	+2.7%	CORRECT
DBLEX	Oct 2010	+2.7%	+2.7%	+0.0%	-2.8%	+1.6%	FALSE
VWOB	Jan 2015	+0.7%	-0.8%	+1.5%	+1.4%	+0.0%	FALSE
MSEDX	Aug 2019	-2.9%	+3.0%	-5.9%	-8.2%	n/a	CORRECT
PFMIX	May 2017	+3.6%	-0.8%	+4.5%	+11.3%	+15.4%	FALSE
PRLAX	Aug 2011	-13.4%	-1.8%	-11.6%	-16.8%	-6.9%	CORRECT
CWGIX	May 2010	+29.4%	+28.7%	+0.7%	+1.0%	+4.6%	FALSE
TEDIX	Nov 2009	+8.6%	+6.6%	+2.0%	+2.5%	+2.6%	FALSE
VHGEX	Oct 2008	+20.8%	+18.6%	+2.2%	+7.2%	+5.7%	FALSE
OPGIX	Oct 2012	+33.7%	+23.4%	+10.3%	+4.0%	+12.4%	FALSE
GPROX	Oct 2018	+13.1%	+12.7%	+0.4%	+22.3%	+47.7%	FALSE
SGIGX	Mar 2020	+56.6%	+54.5%	+2.0%	+2.9%	+0.2%	FALSE
MGGIX	Dec 2008	+77.6%	+32.4%	+45.2%	+80.0%	+65.6%	FALSE

Post-Alarm Analysis

Cumulative excess return vs benchmark in the months following the FIRST alarm — page 5 of 5

Fund	First Alarm	+12M Fund	+12M Bench	+12M Excess	+24M Excess	+36M Excess	12M Signal
AIVSX	May 2019	+9.9%	+7.0%	+2.9%	-0.9%	+3.2%	FALSE
MGIAX	Aug 2018	+2.1%	+0.9%	+1.2%	+4.2%	-8.1%	FALSE
JGVAX	Dec 2019	-7.0%	+15.8%	-22.8%	n/a	n/a	CORRECT
HAINX	May 2013	+15.7%	+16.1%	-0.3%	-7.3%	-16.2%	CORRECT
OP0001I6KI.SI	Nov 2023	+21.6%	+26.5%	-4.9%	-16.6%	n/a	CORRECT
OP00008T6G.SI	Jan 2023	-7.1%	+14.1%	-21.1%	-15.3%	-21.8%	CORRECT
OP0001RMIN.SI	Aug 2024	+7.5%	+16.0%	-8.4%	n/a	n/a	CORRECT
OP0001KL95.SI	Jan 2026	n/a	n/a	n/a	n/a	n/a	FALSE
OP00008T6N.SI	Jan 2023	+3.5%	+5.1%	-1.6%	-7.2%	-12.5%	CORRECT

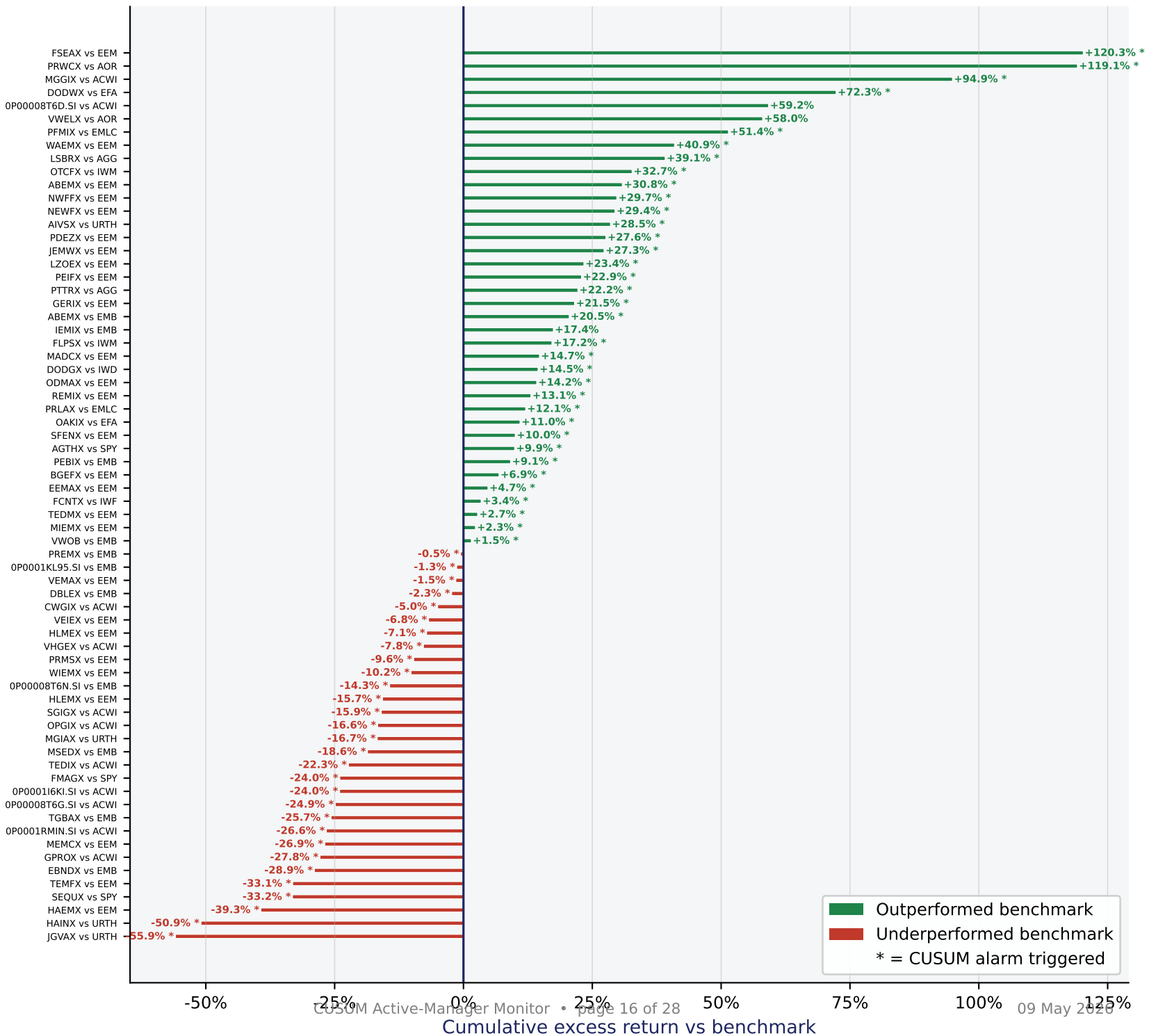
No alarm (no post-alarm analysis): VWELX, IEMIX, OP00008T6D.SI

How to Read This Table

+12M / +24M / +36M Excess: cumulative return of the fund minus the benchmark in the 12, 24, and 36 months immediately AFTER the first alarm.
CORRECT: The fund underperformed its benchmark after the alarm (negative excess) — the CUSUM signal was validated by subsequent returns.
FALSE: The fund outperformed after the alarm — the signal was a false positive. At h=19.81, theory predicts ~1 false alarm per 60 months.

Cumulative Excess Returns — Ranked

Total return of each fund vs its benchmark (Jan 2005 - present)



Individual Fund CUSUM Charts — page 1 of 12

FCNTX vs IWF



DODGX vs IWD



AGTHX vs SPY



OTCFX vs IWM



OAKIX vs EFA

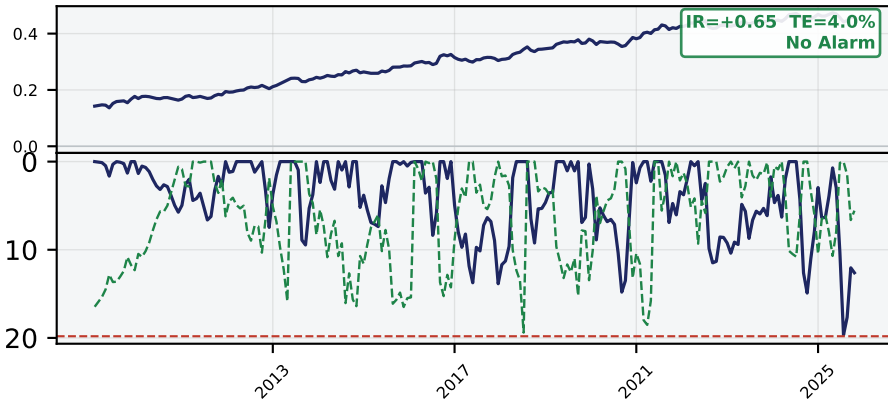


TEDMX vs EEM

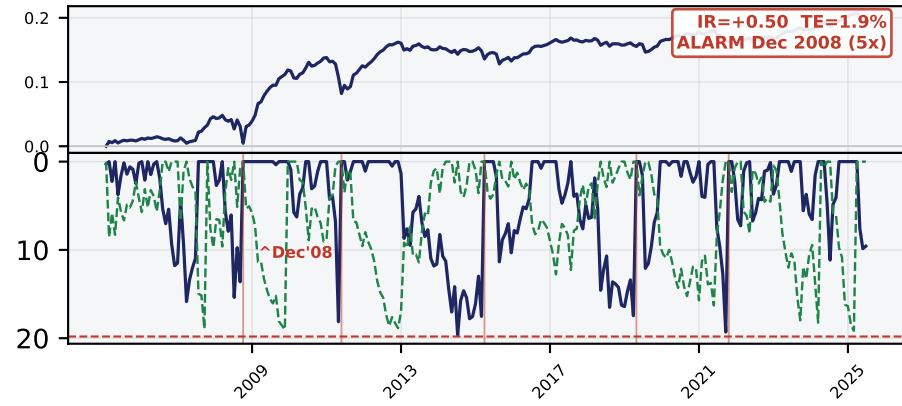


Individual Fund CUSUM Charts — page 2 of 12

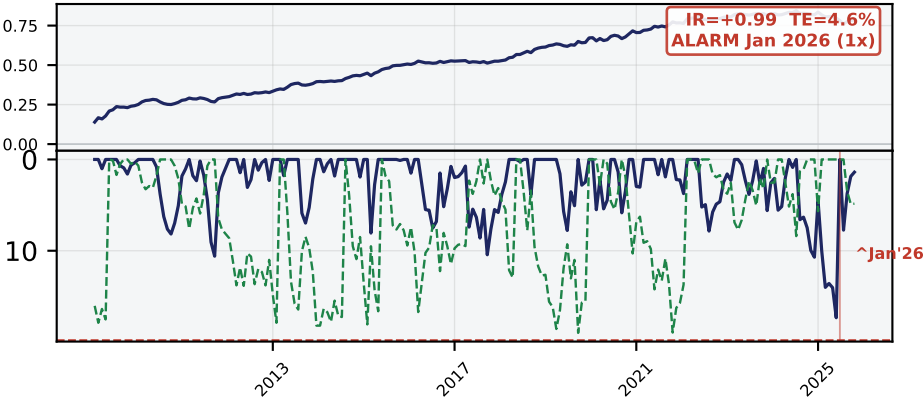
VWELX vs AOR



PTTRX vs AGG



PRWCX vs AOR



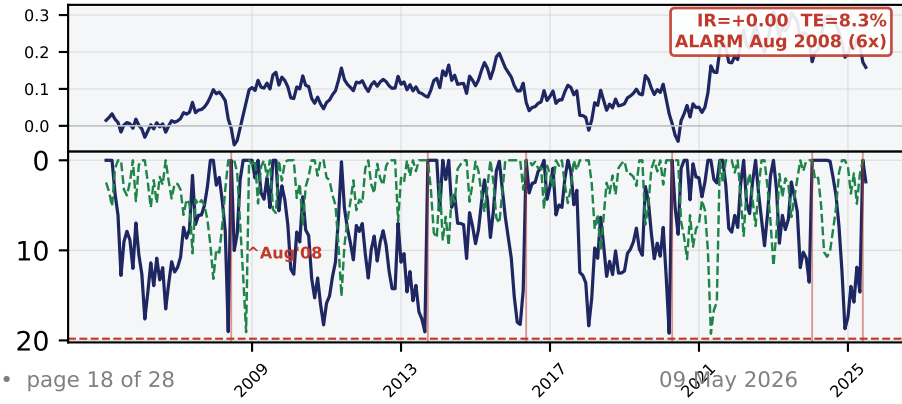
FMAGX vs SPY



SEQUX vs SPY



FLPSX vs IWM



Individual Fund CUSUM Charts — page 3 of 12

DODWX vs EFA



LSBRX vs AGG



NWFFX vs EEM



PRMSX vs EEM



JEMWX vs EEM

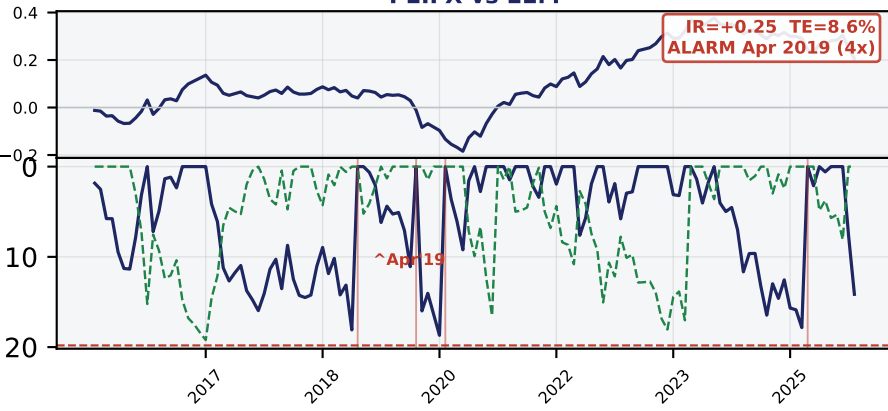


ODMAX vs EEM



Individual Fund CUSUM Charts — page 4 of 12

PEIFX vs EEM



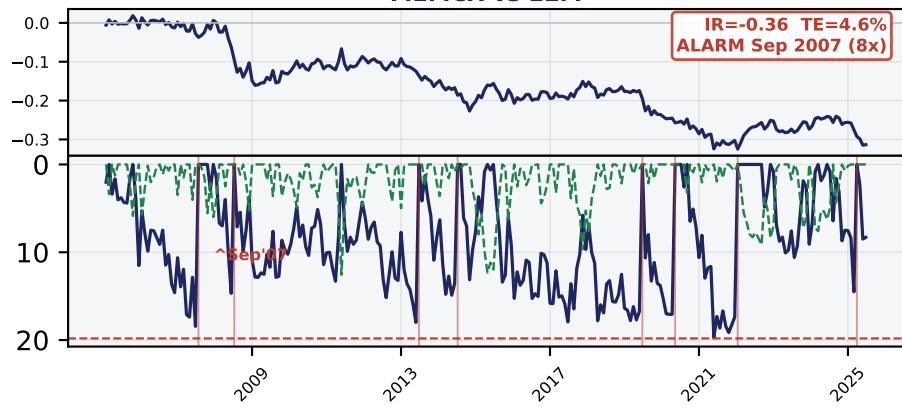
LZOEX vs EEM



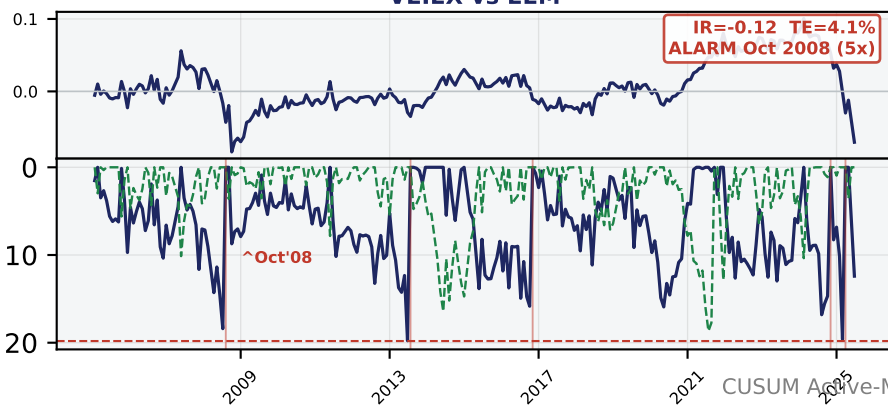
HEMEX vs EEM



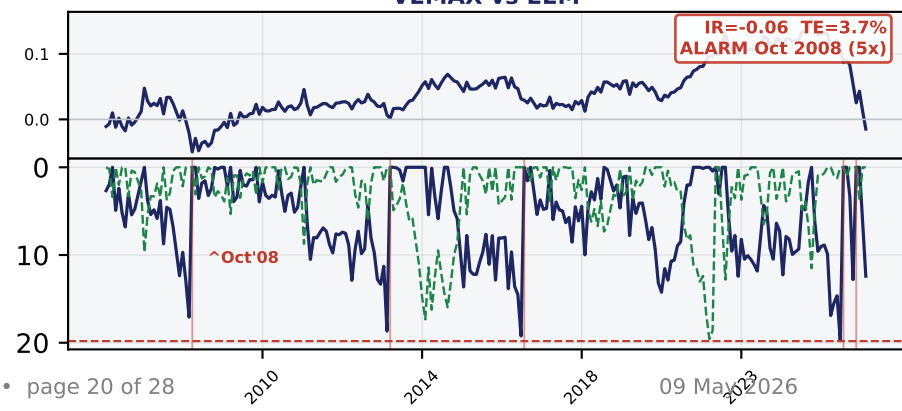
MEMCX vs EEM



VEIEX vs EEM



VEMAX vs EEM



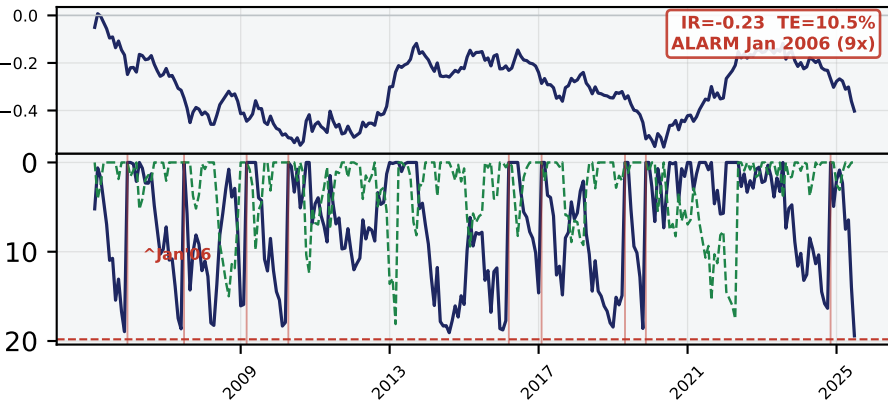
NEWFX vs EEM



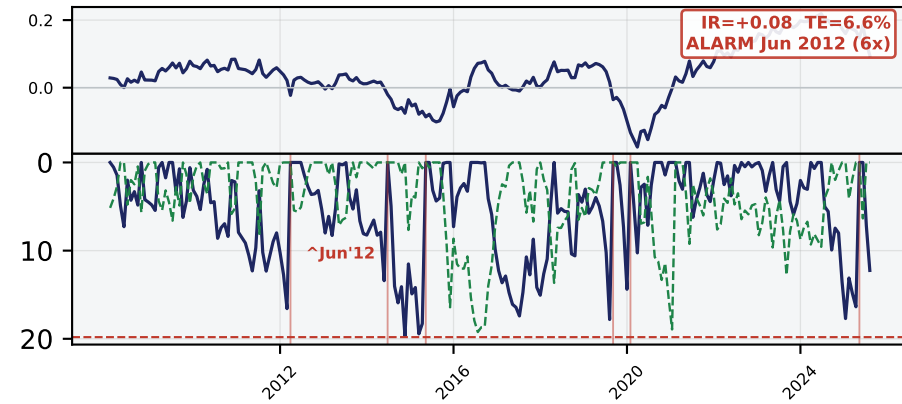
HAEMX vs EEM



TEMFx vs EEM



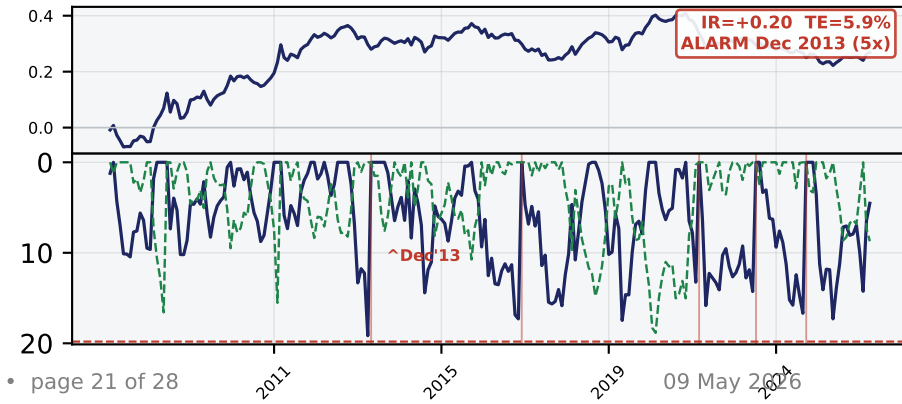
SFENX vs EEM



REMIx vs EEM

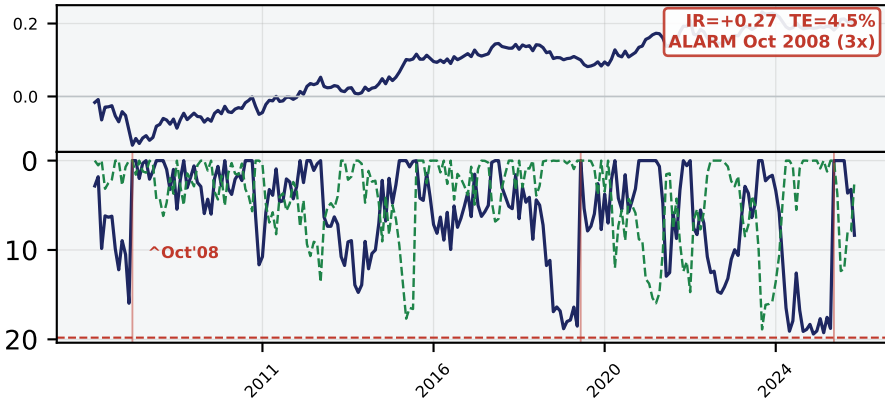


ABEMX vs EEM



Individual Fund CUSUM Charts — page 6 of 12

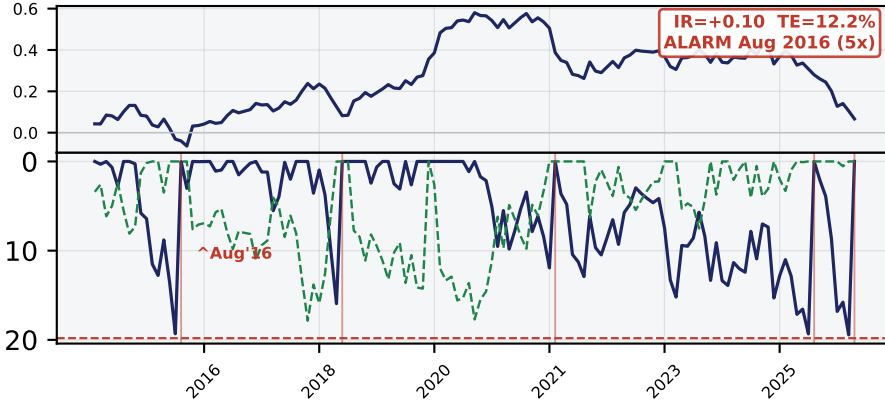
GERIX vs EEM



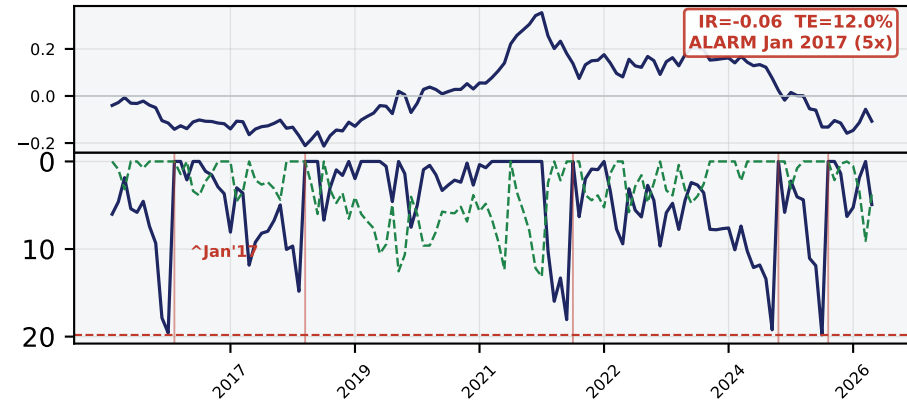
HLEMX vs EEM



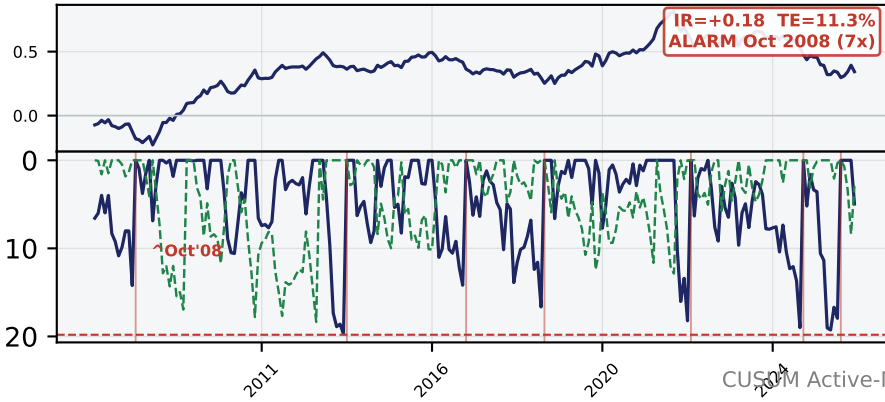
BGEFX vs EEM



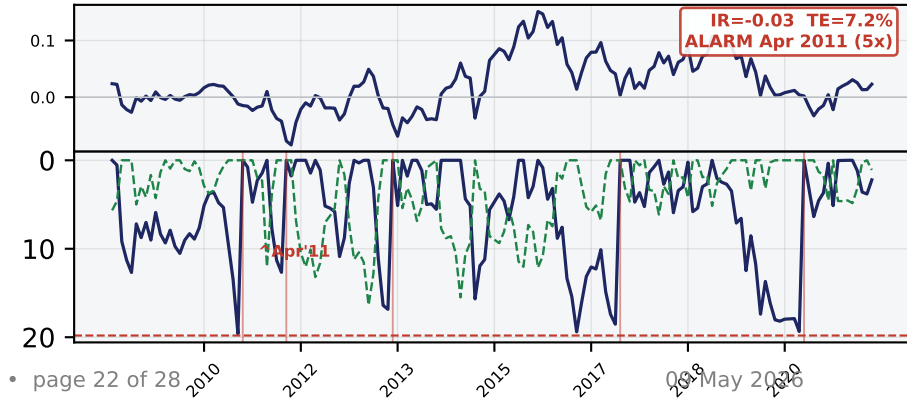
WIEMX vs EEM



WAEMX vs EEM

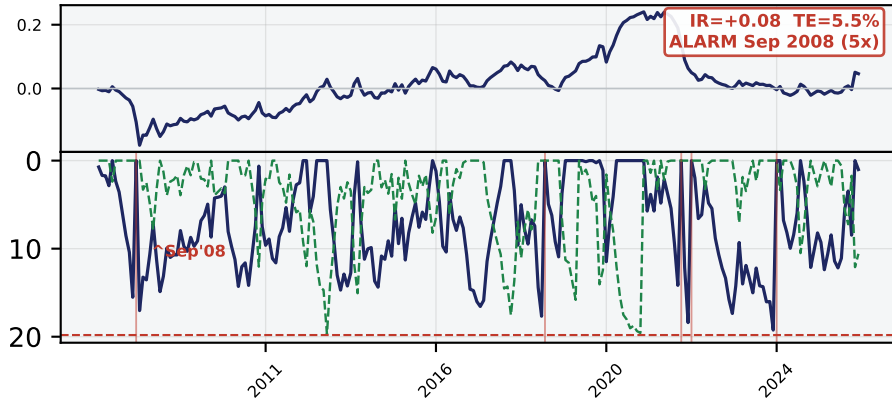


MIEMX vs EEM



Individual Fund CUSUM Charts — page 7 of 12

EEMAX vs EEM



FSEAX vs EEM



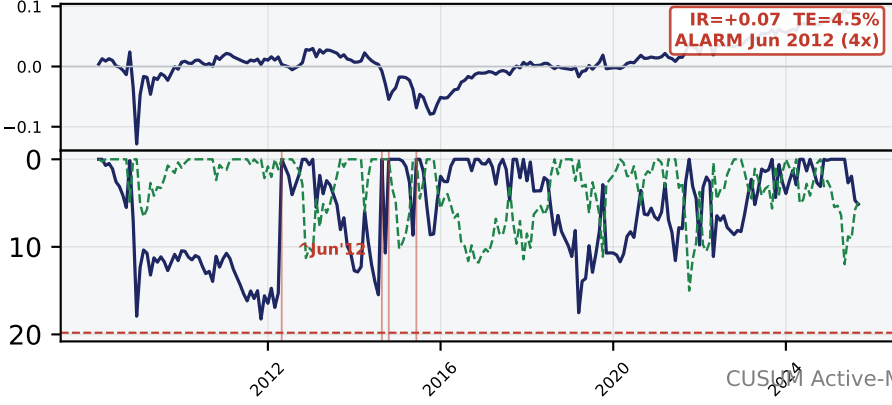
MADCX vs EEM



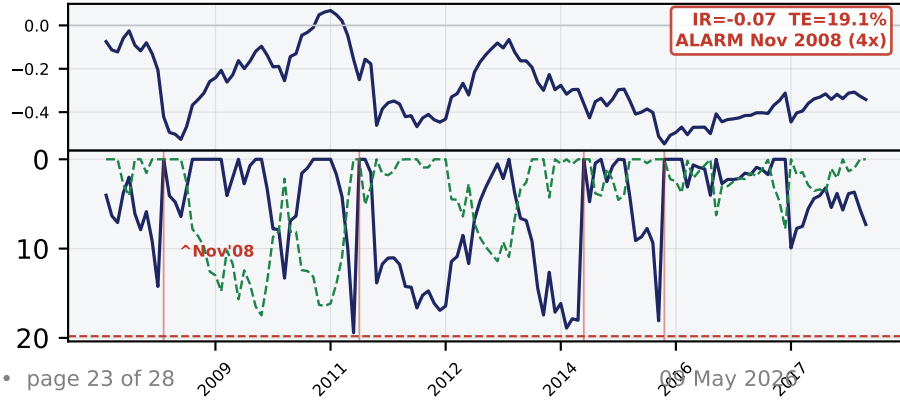
PDEZX vs EEM



PEBIX vs EMB



EBNDX vs EMB



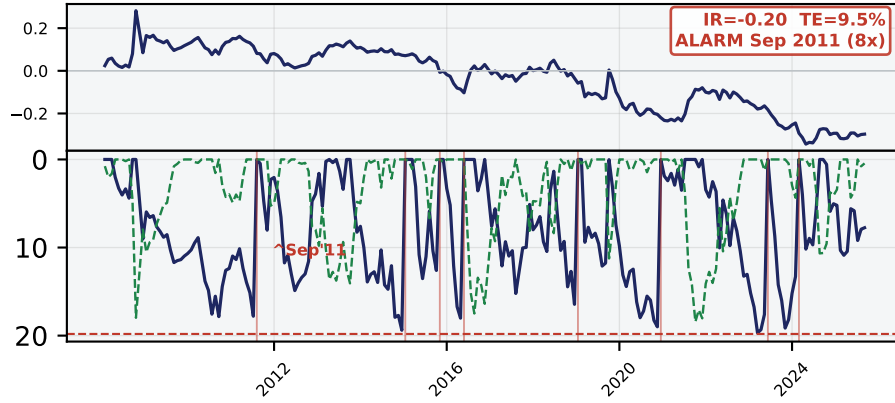
PREMX vs EMB



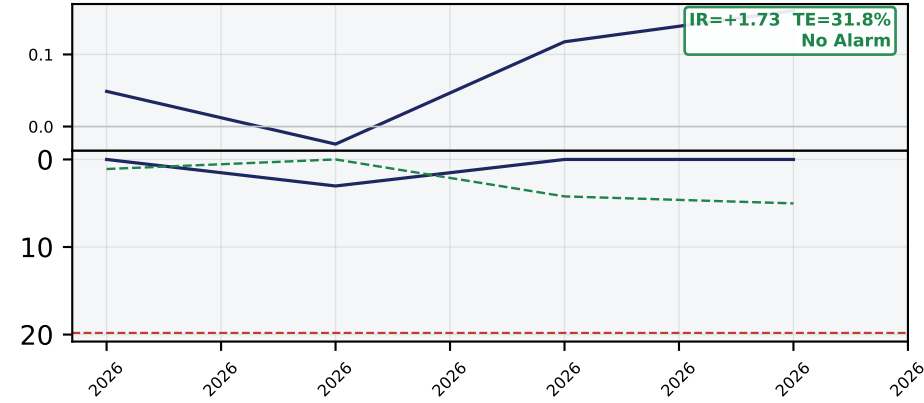
ABEMX vs EMB



TGBAX vs EMB



IEMIX vs EMB



DBLEX vs EMB

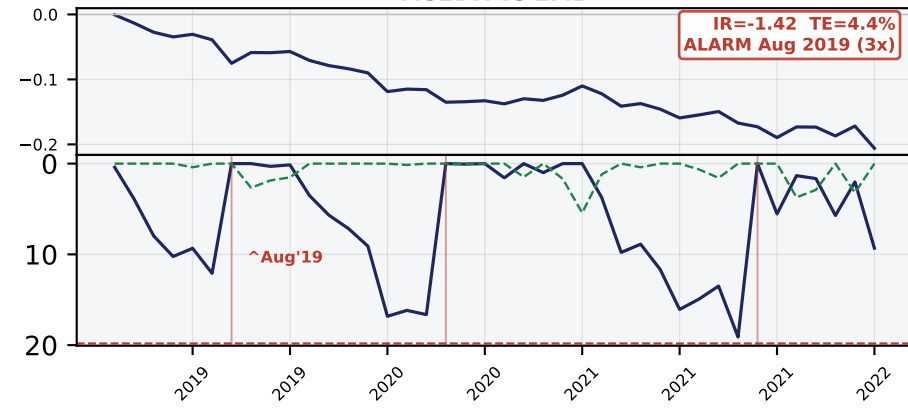


VWOB vs EMB



Individual Fund CUSUM Charts — page 9 of 12

MSEDX vs EMB



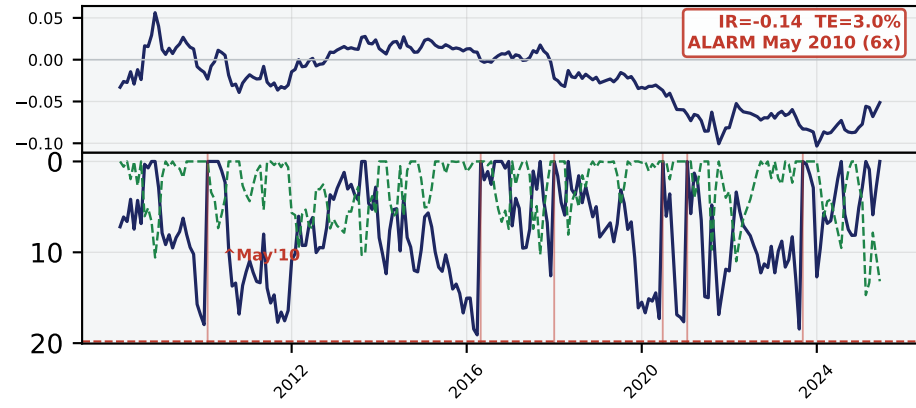
PFMIX vs EMLC



PRLAX vs EMLC



CWGIX vs ACWI



TEDIX vs ACWI



VHGEX vs ACWI



Individual Fund CUSUM Charts — page 10 of 12

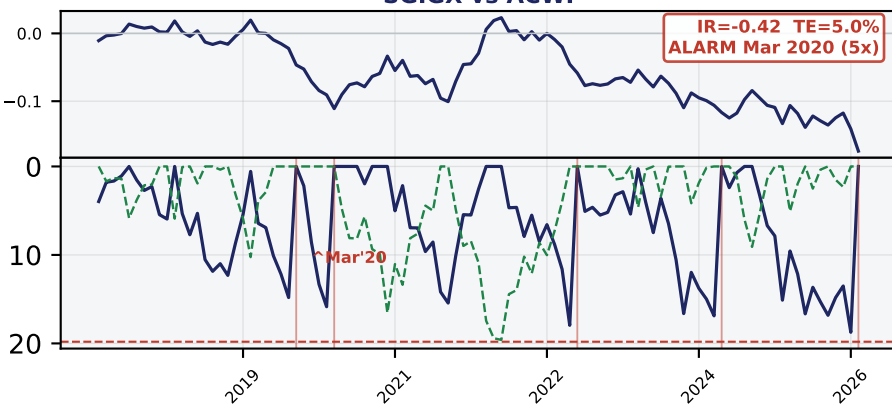
OPGIX vs ACWI



GPROX vs ACWI



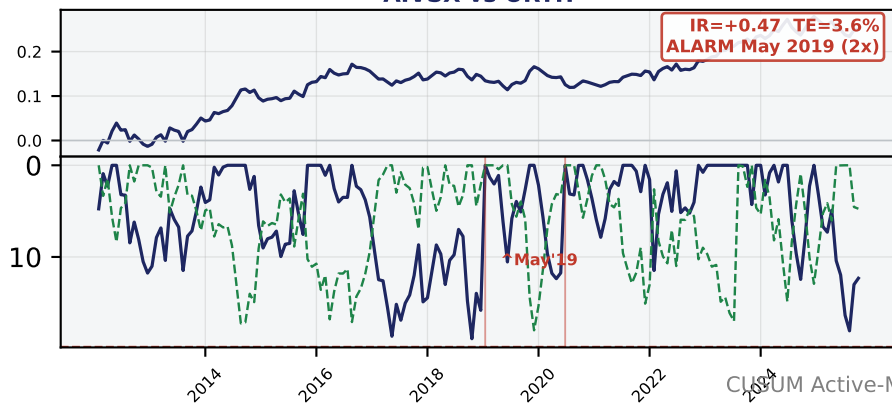
SGIGX vs ACWI



MGGIX vs ACWI



AIVSX vs URT

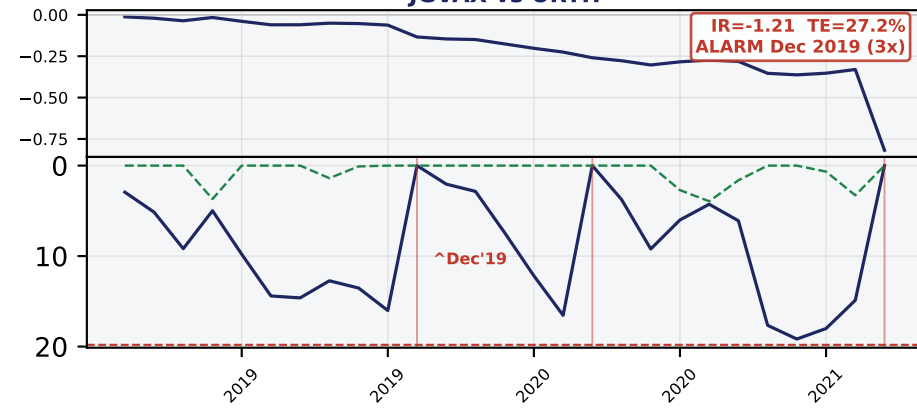


MGIAX vs URT



Individual Fund CUSUM Charts — page 11 of 12

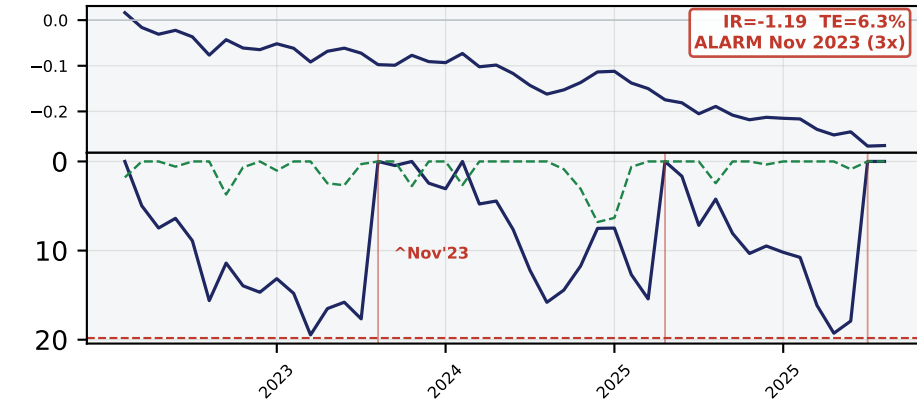
JGVAX vs URTH



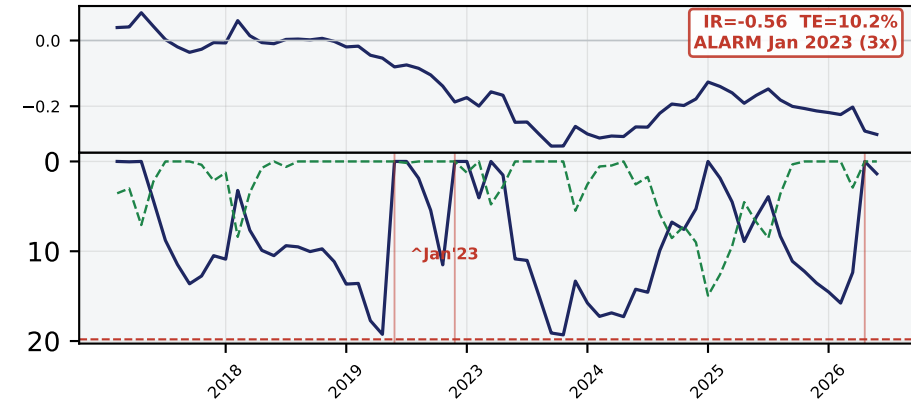
HAINX vs URTH



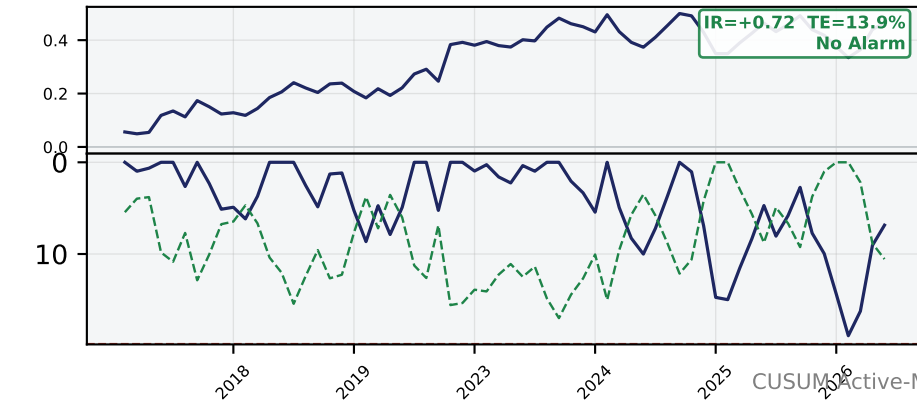
OP0001I6KI.SI vs ACWI



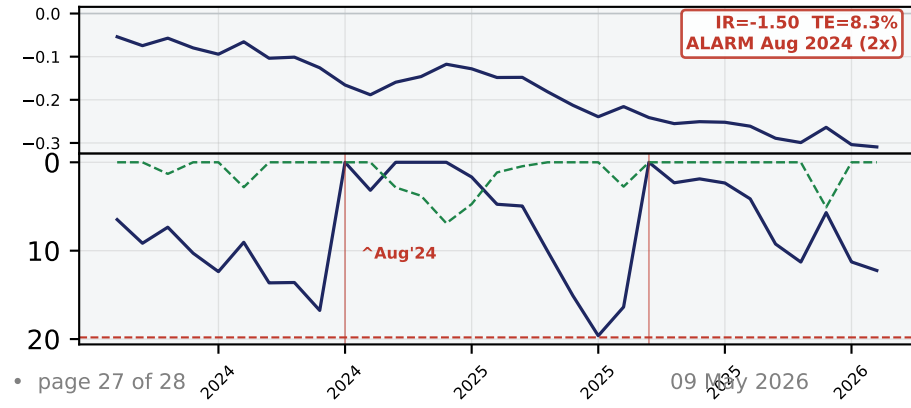
OP00008T6G.SI vs ACWI



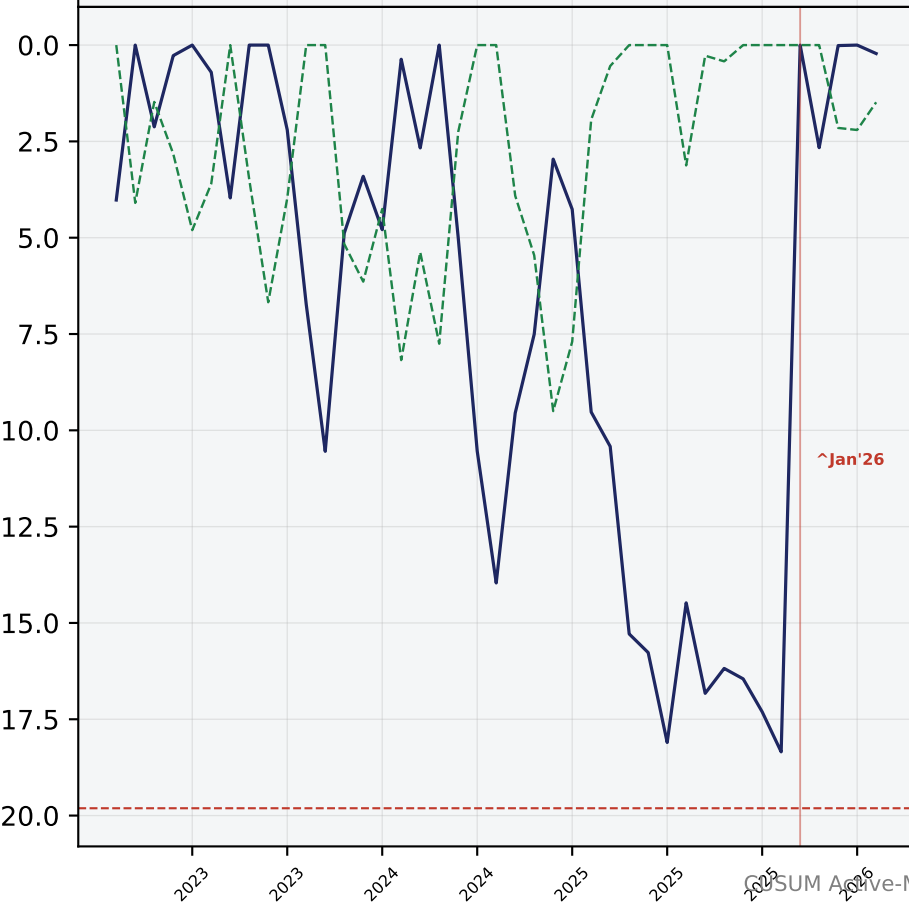
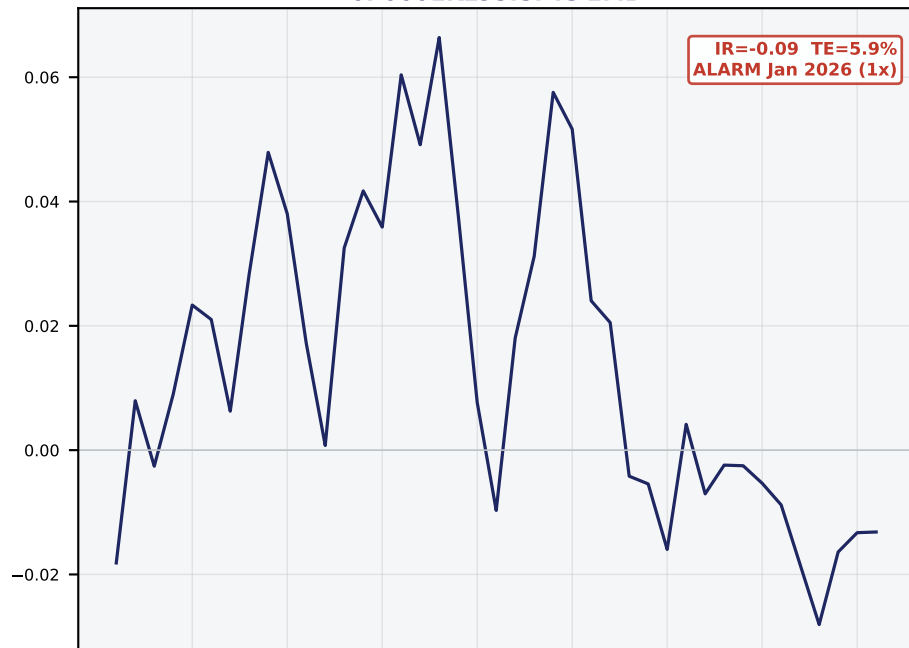
OP00008T6D.SI vs ACWI



OP0001RMIN.SI vs ACWI



OP0001KL95.SI vs EMB



OP00008T6N.SI vs EMB

